

Bessel's Equation & Bessel's Function

The differential equation, $x^2 \frac{d^2 y}{dx^2} + x \frac{dy}{dx} + (x^2 - n^2)y = 0$,

is called the Bessel's differential equation and particular solution of this differential equation are called Bessel's functions of order 'n'.

Solution of Bessel's Differential Equation:

$$x^2 y'' + x y' + (x^2 - n^2)y = 0 \quad \text{--- (1)}$$

$x=0$ is a regular singular point.

$$\text{Let, } y = \sum_{k=0}^{\infty} a_k x^{m+k} \quad \text{--- (2)}$$

$$y' = \sum_{k=0}^{\infty} a_k (m+k) x^{m+k-1}$$

$$y'' = \sum_{k=0}^{\infty} a_k (m+k)(m+k-1) x^{m+k-2}$$

Putting these values in eqⁿ. (1),

$$\begin{aligned} & \sum a_k (m+k)(m+k-1) x^{m+k-2} \cdot x^2 \\ & + \sum a_k (m+k) x^{m+k-1} \cdot x \\ & + \sum a_k x^{m+k} (x^2 - n^2) = 0 \end{aligned}$$

$$\Rightarrow \sum a_k(m+k)(m+k-1)x^{m+k} + \sum a_k(m+k)x^{m+k} + \sum a_k x^{m+k+2} - \sum a_k n^2 x^{m+k} = 0$$

$$\Rightarrow \sum_{k=0}^{\infty} \{(m+k)^2 - n^2\} a_k x^{m+k} + \sum_{k=0}^{\infty} a_k x^{m+k+2} = 0$$

The lowest power of x is x^m when k is zero....

Equating to zero the coefficient of x^m , we get,

$$m^2 - n^2 = 0 \quad \text{as } a_0 \neq 0 \quad [k=0]$$

$$\Rightarrow m = \pm n$$

Equating to zero the coefficient of next term i.e. x^{m+1} , we get,

$$[(m+1)^2 - n^2] a_1 = 0 \quad [k=1]$$

$$\Rightarrow a_1 = 0$$

Equating to zero the coefficient of x^{m+k+2} ,

$$a_{k+2} = \frac{-a_k}{(m+k+2)^2 - n^2} \quad (3) \quad [k=k+2]$$

$$[\{(m+k+2)^2 - n^2\} a_{k+2} + a_k] x^{m+k+2} = 0$$

Putting $k=1$ in eqⁿ. (3),

$$a_{1+2} = a_3 = \frac{-a_1}{(m+3)^2 - n^2} = 0$$

Similarly, $a_5 = a_7 = \dots = 0$ for $k=1, 3, 5, \dots$

Putting $k=0$ in eqⁿ. (3),

$$a_{0+2} = a_2 = \frac{-a_0}{(m+2)^2 - n^2}$$

Similarly, $a_4 = \frac{-a_2}{(m+4)^2 - n^2} \quad [k=2]$

$$\text{i.e. } a_4 = \frac{a_0}{[(m+2)^2 - n^2][(m+4)^2 - n^2]}$$

So, for $k=0, 2, 4, \dots$

$$a_2 = \frac{-a_0}{(m+2)^2 - n^2}, \quad a_4 = \frac{a_0}{[(m+2)^2 - n^2][(m+4)^2 - n^2]}, \dots$$

Now, eqⁿ. (2) becomes,

$$y = a_0 x^m \left[1 - \frac{x^2}{(m+2)^2 - n^2} + \frac{x^4}{[(m+2)^2 - n^2][(m+4)^2 - n^2]} + \dots \right]$$

general series (4)

This is the ~~solution~~ solution of Bessel's Equation.

Case-I: ~~When~~ When n is not an integer.

For $m=n$,

$$y_1 = a_0 x^n \left[1 - \frac{x^2}{2(2n+2)} + \frac{x^4}{2 \cdot 4 \cdot (2n+2)(2n+4)} - \dots \right]$$

$$\Rightarrow y_1 = a_0 x^n \left[1 + \frac{(-1)^1 x^2}{2^2 \cdot 1! (n+1)} + \frac{(-1)^2 x^4}{2^4 \cdot 2! (n+1)(n+2)} + \dots \right]$$

$$\Rightarrow y_1 = a_0 x^n \sum_{k=0}^{\infty} \frac{(-1)^k \sqrt{n+1}}{2^{2k} k! \sqrt{n+k+1}} \cdot x^{2k} \quad \text{--- (5)}$$

For $m=-n$,

$$y_2 = a_0 x^{-n} \sum_{k=0}^{\infty} \frac{(-1)^k \sqrt{-n+1}}{2^{2k} k! \sqrt{-n+k+1}} \cdot x^{2k} \quad \text{--- (6)}$$

The complete solution of differential eqⁿ. (1) is,

$$y = C_1 y_1 + C_2 y_2 \quad \text{--- (7)}$$

Since, a_0 is arbitrary ~~so~~ so if taken $a_0 = \frac{1}{2^n \sqrt{n+1}}$

in the solution (5), then this solution is called Bessel's function of first kind and is denoted by $J_n(x)$.

$$J_n(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \sqrt{n+k+1}} \left(\frac{x}{2}\right)^{n+2k} \quad \text{--- (8)}$$

Similarly, if taking $a_0 = \frac{1}{2^{-n} \sqrt{-n+1}}$ in (6), then, this solution is called Bessel's function of second kind and denoted by $J_{-n}(x)$.

$$J_{-n}(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \sqrt{-n+k+1}} \left(\frac{x}{2}\right)^{-n+2k} \quad \text{--- (9)}$$

Hence, the general solution of eqⁿ. (1) when 'n' is not an integer is,

$$y = A \cdot J_n(x) + B \cdot J_{-n}(x) \quad \text{--- (10)}$$

where, A & B are the arbitrary constants.

Case-II: When $n = 0$.

The Bessel's differential eqⁿ. (1) becomes,

$$x^2 y'' + x y' + x^2 y = 0 \quad \text{--- (11)}$$

This eqⁿ. (11) is known as the Bessel's Equation of Order Zero. In this case, the roots of the indicial equation are equal.

$$\text{i.e. } m = 0, 0$$

For $m = 0$, the first solution is the Bessel's function of the first kind of order zero.

$$J_0(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(k!)^2} \left(\frac{x}{2}\right)^{2k} \quad \text{--- (12)} \\ [\because k! = k!]$$

To find the second independent solution when roots are equal, we use the derivative w.r.to the 'm',

$$\frac{\partial y}{\partial m} = x^m \log x \left[1 - \frac{x^2}{(m+2)^2} + \frac{x^4}{(m+2)^2(m+4)^2} - \dots \right] \\ + x^m \left[\frac{x^2}{(m+2)^2} \cdot \frac{2}{m+2} - \frac{x^4}{(m+2)^2(m+4)^2} \left(\frac{2}{m+2} + \frac{2}{m+4} \right) + \dots \right]$$

[$a_0 = 1$, $n = 0$ & applying $d(uv)$]

$$\Rightarrow \left(\frac{dy}{dm} \right)_{m=0} = J_0(x) \cdot \log x + \left[\left(\frac{x}{2} \right)^2 - \frac{1}{2^2 \cdot 4^2} \left(1 + \frac{1}{2} \right) x^4 + \frac{1}{2^2 \cdot 4^2 \cdot 6^2} \left(1 + \frac{1}{2} + \frac{1}{3} \right) x^6 - \dots \right]$$

$$\Rightarrow Y_0(x) = J_0(x) \cdot \log x + \left(\frac{x}{2} \right)^2 - \frac{1}{(2!)^2} \left(1 + \frac{1}{2} \right) \left(\frac{x}{2} \right)^4 + \frac{1}{(3!)^2} \left(1 + \frac{1}{2} + \frac{1}{3} \right) \left(\frac{x}{2} \right)^6 - \dots$$

$$\Rightarrow Y_0(x) = J_0(x) \cdot \log x + \sum_{k=1}^{\infty} \frac{(-1)^{k+1}}{(k!)^2} \left(1 + \frac{1}{2} + \dots + \frac{1}{k} \right) \left(\frac{x}{2} \right)^{2k} \quad (13)$$

This is known as the Bessel function of the second kind of order zero - (or the Neumann-function).

Hence, the complete general solution for the Bessel equation of order zero is,

$$y = A \cdot J_0(x) + B \cdot Y_0(x) \quad (14)$$

where, A & B are arbitrary constants.

Case-III: When n is an integer.

In this case, the two functions $J_n(x)$ & $J_{-n}(x)$ are not independent but are connected by a relation,

$$J_{-n}(x) = (-1)^n J_n(x) \text{ ————— (15)}$$

Let us find an independent solution of Bessel's equation (1) when n is an integer.

Let, $y = u(x) \cdot J_n(x)$ — (16), be a solution of eqⁿ. (1).

Then,

$$y' = u' J_n + u J_n'$$

$$\& \quad y'' = u'' J_n + 2u' J_n' + u J_n''$$

Putting these values in eqⁿ. (1), we get,

$$x^2(u'' J_n + 2u' J_n' + u J_n'') + x(u' J_n + u J_n') + (x^2 - n^2) u J_n = 0$$

$$\Rightarrow u[x^2 J_n'' + x J_n' + (x^2 - n^2) J_n] + x^2 u'' J_n + 2x^2 u' J_n' + x u' J_n = 0$$

$$\Rightarrow u \times 0 + x^2 u'' J_n + 2x^2 u' J_n' + x u' J_n = 0$$

[$\because J_n$ is a solution of eqⁿ. (1)]

$$x^2 u'' J_n + 2x^2 u' J_n' + x u' J_n = 0$$

Dividing by $x^2 u' J_n$, we get,

$$\frac{u''}{u'} + 2 \frac{J_n'}{J_n} + \frac{1}{x} = 0$$

On Integrating w.r.to x , we get,

$$\log(u' J_n^2 x) = \log B \quad \left[\int \frac{f'(x)}{f(x)} dx = \log|f(x)| \right]$$

$$\Rightarrow u' J_n^2 x = B$$

$$\Rightarrow u' = \frac{B}{x J_n^2}$$

$$\Rightarrow u = B \int \frac{dx}{x J_n^2} + A$$

Putting this value of u in eqⁿ. (16),

$$y = \left[B \int \frac{dx}{x J_n^2(x)} + A \right] \cdot J_n(x)$$

$$\Rightarrow y = A J_n(x) + B Y_n(x) \quad \text{--- (17)}$$

where, the function $Y_n(x)$ is called the ~~the~~ Bessel function of the second kind of order n or Neumann function.

Series Representation of Bessel's Function:

$$\text{Since, } J_n(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(n+k+1)} \left(\frac{x}{2}\right)^{n+2k}$$

Taking $n=0$,

$$J_0(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{k! \Gamma(k+1)} \left(\frac{x}{2}\right)^{2k}$$

$$\Rightarrow J_0(x) = \sum_{k=0}^{\infty} \frac{(-1)^k}{(k!)^2} \left(\frac{x}{2}\right)^{2k}$$

$$\Rightarrow J_0(x) = 1 - \left(\frac{x}{2}\right)^2 + \frac{1}{(2!)^2} \left(\frac{x}{2}\right)^4 - \frac{1}{(3!)^2} \left(\frac{x}{2}\right)^6 + \dots$$

$$\Rightarrow J_0(x) = 1 - \frac{x^2}{2^2} + \frac{x^4}{2^2 \cdot 4^2} - \frac{x^6}{2^2 \cdot 4^2 \cdot 6^2} + \dots$$

Similarly, if $n=1$,

$$J_1(x) = \frac{x}{2} - \frac{x^3}{2^2 \cdot 4} + \frac{x^5}{2^2 \cdot 4^2 \cdot 6} - \dots$$

In particular, $J_0(0) = 1$ & $J_1(0) = 0$.

Recurrence Relations for $J_n(x)$:

1. $x J_n' = n J_n - x J_{n+1}$

Proof:

$$J_n(x) = \sum_{r=0}^{\infty} \frac{(-1)^r}{r! \sqrt{n+r+1}} \left(\frac{x}{2}\right)^{n+2r}$$

Diff. w.r. to x , we get,

$$J_n'(x) = \sum_{r=0}^{\infty} \frac{(-1)^r (n+2r)}{r! \sqrt{n+r+1}} \cdot \frac{1}{2} \cdot \left(\frac{x}{2}\right)^{n+2r-1}$$

Multiplying both sides by x and splitting the terms,

$$x J_n'(x) = n \sum_{r=0}^{\infty} \frac{(-1)^r}{r! \sqrt{n+r+1}} \left(\frac{x}{2}\right)^{n+2r}$$

$$+ \sum_{r=0}^{\infty} \frac{(-1)^r (2r)}{r! \sqrt{n+r+1}} \left(\frac{x}{2}\right)^{n+2r}$$

$$\Rightarrow x J_n'(x) = n J_n(x) + \sum_{r=1}^{\infty} \frac{(-1)^r (2r)}{r(r-1)! \sqrt{n+r+1}} \left(\frac{x}{2}\right)^{n+2r}$$

$$\Rightarrow x J_n'(x) = n J_n(x) + \sum_{r=1}^{\infty} \frac{(-1)^r \cdot 2}{(r-1)! \sqrt{n+r+1}} \left(\frac{x}{2}\right)^{n+2r}$$

$$\Rightarrow x J_n'(x) = n J_n(x) + \sum_{r=1}^{\infty} \frac{(-1)^r \cdot 2}{(r-1)! \sqrt{n+r+1}} \left(\frac{x}{2}\right)^{n+2r-1}$$

$$\Rightarrow x J_n'(x) = n J_n(x) + x \sum_{r=1}^{\infty} \frac{(-1)^r}{(r-1)! \sqrt{n+r+1}} \left(\frac{x}{2}\right)^{n+2r-1}$$

Let, $r-1 = s$
 $\Rightarrow r = s+1$

$$x J_n'(x) = n J_n(x) + x \sum_{s=1}^{\infty} \frac{(-1)^{s+1}}{s! \sqrt{n+s+2}} \left(\frac{x}{2}\right)^{n+2s+2-1}$$

$$\Rightarrow x J_n'(x) = n J_n(x) - x \sum_{s=1}^{\infty} \frac{(-1)^s}{s! \sqrt{(n+1)+s+1}} \left(\frac{x}{2}\right)^{(n+1)+2s}$$

$$\Rightarrow x J_n'(x) = n J_n(x) - x J_{n+1}(x)$$

i.e. $\boxed{x J_n' = n J_n - x J_{n+1}}$. proved

$$2. \quad x J_n' = -n J_n + x J_{n-1}$$

Proof:

$$J_n(x) = \sum_{r=0}^{\infty} \frac{(-1)^r}{r! \sqrt{n+r+1}} \cdot \left(\frac{x}{2}\right)^{n+2r}$$

$$\Rightarrow J_n' = \sum_{r=0}^{\infty} \frac{(-1)^r (n+2r)}{r! \sqrt{n+r+1}} \cdot \frac{1}{2} \cdot \left(\frac{x}{2}\right)^{n+2r-1}$$

$$\Rightarrow x J_n' = \sum_{r=0}^{\infty} \frac{(-1)^r (2n+2r-n)}{r! \sqrt{n+r+1}} \cdot \left(\frac{x}{2}\right)^{n+2r}$$

$$\Rightarrow x J_n' = -n \sum_{r=0}^{\infty} \frac{(-1)^r}{r! \sqrt{n+r+1}} \cdot \left(\frac{x}{2}\right)^{n+2r}$$

$$+ \sum_{r=0}^{\infty} \frac{(-1)^r (2n+2r)}{r! \sqrt{n+r+1}} \cdot \left(\frac{x}{2}\right)^{n+2r}$$

$$\Rightarrow x J_n' = -n J_n + x \sum_{r=0}^{\infty} \frac{(-1)^r (n+r)}{r! \sqrt{n+r+1}} \cdot \left(\frac{x}{2}\right)^{n+2r-1}$$

$$\Rightarrow x J_n' = -n J_n + x \sum_{r=0}^{\infty} \frac{(-1)^r (n+r)}{r! (n+r) \sqrt{n+r}} \cdot \left(\frac{x}{2}\right)^{n+2r-1}$$

$$[\because \sqrt{n+1} = n \sqrt{n}]$$

$$\Rightarrow x J_n' = -n J_n + x \sum_{r=0}^{\infty} \frac{(-1)^r}{r! \sqrt{(n-1)+r+1}} \cdot \left(\frac{x}{2}\right)^{(n-1)+2r}$$

$$\Rightarrow \boxed{x J_n' = -n J_n + x J_{n-1}} \quad \text{proved}$$

3. $2 J_n' = J_{n-1} - J_{n+1}$

Proof:

Adding (1) & (2),

$$x J_n' + x J_n' = \cancel{n J_n} - x J_{n+1} - \cancel{n J_n} + x J_{n-1}$$

$$\Rightarrow 2 J_n' = J_{n-1} - J_{n+1}.$$

4. $2n J_n = x(J_{n-1} + J_{n+1})$

Proof:

Subtracting (2) from (1), we get,

$$0 = 2n J_n - x J_{n-1} - x J_{n+1}$$

$$\Rightarrow 2n J_n = x(J_{n-1} + J_{n+1}).$$

5. $\frac{d}{dx} (x^{-n} J_n) = -x^{-n} J_{n+1}$

Proof:

Multiplying (1) by x^{-n-1} , we get,

$$x^{-n} J_n' = n x^{-n-1} J_n - x^{-n} J_{n+1}$$

$$\Rightarrow x^{-n} J_n' - x^{-n-1} n J_n = -x^{-n} J_{n+1}$$

$$\Rightarrow \frac{d}{dx} (x^{-n} J_n) = -x^{-n} J_{n+1}$$

$$\left[\frac{d}{dx} (x^{-n} J_n) = x^{-n} J_n' - x^{-n-1} n J_n \right] \text{ (Product Rule)}$$

6. $\frac{d}{dx} (x^n J_n) = x^n J_{n-1}$

Proof:

Multiplying (2) by x^{n+1} , we get,

$$x^n J_n' = -n x^{n-1} J_n + x^n J_{n-1}$$

$$\Rightarrow x^n J_n' + n x^{n-1} J_n = x^n J_{n-1}$$

$$\Rightarrow \frac{d}{dx} (x^n J_n) = x^n J_{n-1}$$

Questions

Imp. 1. Prove that: $J_{-n}(x) = (-1)^n J_n(x)$, where 'n' is a positive integer.

Solⁿ:

$$J_{-n}(x) = \sum_{\gamma=0}^{\infty} \frac{(-1)^{\gamma}}{\gamma! \sqrt{-n+\gamma+1}} \left(\frac{x}{2}\right)^{n+2\gamma}$$

$$= \sum_{\gamma=0}^{n-1} \frac{(-1)^{\gamma}}{\gamma! \sqrt{-n+\gamma+1}} \left(\frac{x}{2}\right)^{-n+2\gamma}$$

$$+ \sum_{\gamma=n}^{\infty} \frac{(-1)^{\gamma}}{\gamma! \sqrt{-n+\gamma+1}} \left(\frac{x}{2}\right)^{-n+2\gamma}$$

$$= 0 + \sum_{s=0}^{\infty} \frac{(-1)^{n+s}}{(n+s)! \sqrt{-n+(n+s)+1}} \left(\frac{x}{2}\right)^{-n+2(n+s)}$$

For the first summation $(-n+\gamma+1) \leq 0$, so,
 $\frac{1}{\sqrt{-p}} = \frac{1}{\infty} = 0$.

For the second summation, we put, $\gamma = n+s$.
When $\gamma = n$, $s = 0$.

$$J_{-n}(x) = \sum_{s=0}^{\infty} \frac{(-1)^n \cdot (-1)^s}{(n+s)! \sqrt{s+1}} \cdot \left(\frac{x}{2}\right)^{n+2s}$$

$$\Rightarrow J_{-n}(x) = (-1)^n \sum_{s=0}^{\infty} \frac{(-1)^s}{s! \sqrt{n+s+1}} \left(\frac{x}{2}\right)^{n+2s}$$

$$\left[\begin{array}{l} \sqrt{s+1} = s! \\ \& (n+s)! = \sqrt{n+s+1} \end{array} \right]$$

$$\Rightarrow J_{-n}(x) = (-1)^n J_n(x) \quad \underline{\text{proved}}$$

2. Prove that: $J_0'(x) = -J_1(x)$.

Solⁿ:

We have,

$$\frac{d}{dx} (x^{-n} J_n(x)) = -x^{-n} J_{n+1}(x)$$

Putting $n=0$,

$$\frac{d}{dx} [J_0(x)] = -J_1(x)$$

$$\Rightarrow J_0'(x) = -J_1(x) \quad \underline{\text{proved}}$$

3. Prove that:

$$(i) J_{1/2}(x) = \frac{\sqrt{2}}{\sqrt{\pi x}} \sin x \quad \& \quad (ii) J_{-1/2}(x) = \frac{\sqrt{2}}{\sqrt{\pi x}} \cos x$$

Solⁿ:

$$J_n(x) = \sum_{r=0}^{\infty} \frac{(-1)^r}{r! \Gamma(n+r+1)} \left(\frac{x}{2}\right)^{n+2r}$$

$$\Rightarrow J_n(x) = \frac{x^n}{2^n \Gamma(n+1)} \left[1 - \frac{x^2}{2(2n+2)} + \frac{x^4}{2 \cdot 4 \cdot (2n+2)(2n+4)} - \dots \right] \quad (1)$$

(i) Putting $n = 1/2$ in eqⁿ. (1),

$$J_{1/2}(x) = \frac{x^{1/2}}{2^{1/2} \Gamma(3/2)} \left[1 - \frac{x^2}{2 \cdot 2(1/2+1)} + \frac{x^4}{2^3 \cdot 4(1/2+1)(1/2+2)} - \dots \right]$$

~~$$\Rightarrow J_{1/2}(x) = \frac{\sqrt{x}}{\sqrt{2} \cdot 1/2 \Gamma(1/2)} \cdot \frac{1}{x} \left[x - \frac{x^3}{3} + \dots \right]$$~~

$$\Rightarrow J_{1/2}(x) = \frac{\sqrt{x}}{\sqrt{2} \Gamma(3/2)} \left[1 - \frac{x^2}{2 \cdot 3} + \frac{x^4}{2 \cdot 3 \cdot 4 \cdot 5} - \dots \right]$$

$$\Rightarrow J_{1/2}(x) = \frac{\sqrt{x}}{\sqrt{2} \cdot 1/2 \Gamma(1/2)} \cdot \frac{1}{x} \left[x - \frac{x^3}{3!} + \frac{x^5}{5!} - \dots \right]$$

$$\Rightarrow J_{1/2}(x) = \frac{1}{\sqrt{2x} \cdot 1/2 \sqrt{\pi}} \sin x$$

$$\Rightarrow J_{1/2}(x) = \frac{\sqrt{2}}{\sqrt{\pi x}} \sin x \quad \text{proved}$$